

Youssef Ahmed

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SUMMARY

Software Engineer with experience building full-stack web applications, AI-powered platforms, and automated testing solutions using React, Node.js, MongoDB, Java, Selenium, and OpenAI APIs. Strong foundation in backend development, REST APIs, automated testing, performance optimization, and software design.

EDUCATION

BSc Computer Science — Software Engineering — The British University in Egypt (BUE), affiliated with London South Bank University | Sep 2021 – Sep 2026

CERTIFICATIONS

Interactive Programming in Python (Rice University) • Computational Thinking for Problem Solving (UPenn) • Business Intelligence and Data Warehousing (UNAM) — Coursera, May 2026

PROJECTS

ComicMind — AI Comic Generation Platform — React, TypeScript, Node.js, Express, MongoDB, OpenAI API, Cloudinary | 2025–2026

- Achieved 100% user satisfaction on generated image quality and 100% recommendation rate in a structured user evaluation study by architecting an end-to-end AI pipeline with deterministic character-consistency prompting and OpenAI GPT-Image integration.
- Delivered 0% failed requests and ~118ms average API response time under concurrent load (K6 benchmark) by building and load-testing a Node.js/Express REST API backed by MongoDB Atlas.
- Cut comic-retrieval query cost by adding targeted MongoDB indexes that shifted query execution from COLLSCAN to IXSCAN.
- Verified authentication and CRUD reliability with Vitest and Supertest tests covering registration, login, and comic management endpoints.

Glassdoor Web Application — Automated Testing — Java, Selenium WebDriver, TestNG, Cucumber, Gherkin | Software Testing Project

- Developed automated UI tests for Glassdoor using Selenium WebDriver and Java, with TestNG for test execution.
- Created positive and negative automated scenarios for job search and company reviews, and tested employer overview functionality.
- Implemented BDD scenarios using Cucumber/Gherkin and structured test cases with preconditions, test data, expected results, and alternative paths.

Game Discovery Platform — React, Node.js, Express, MongoDB, JWT, RAWG API, Vercel | 2025

- Delivered a full-stack platform spanning game browsing/search, authentication, favorites, reviews, and an admin dashboard using a JWT-secured backend and responsive React frontend.

Online Auction System — Java, Swing, JUnit5 | 2025

- Automated core bidding and auction lifecycle tests using JUnit5 while applying Strategy, Observer, and Template Method design patterns.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, C++, SQL, HTML, CSS

Frontend: React, TypeScript, Tailwind CSS, Vite

Backend & Databases: Node.js, Express, MongoDB, MongoDB Atlas, REST APIs, JWT

Testing & Performance: Selenium WebDriver, TestNG, Cucumber, Gherkin, Vitest, Supertest, JUnit5, K6

Tools & Platforms: Git, GitHub, Vercel, Cloudinary, VS Code, Postman

AI: OpenAI API (GPT-4, GPT-Image), GitHub Copilot, Prompt Engineering

LANGUAGES

Arabic: Native | English: Fluent